

The Industrial Revolution

In mid-18th-century Britain, a surge in new technology and inventions meant that fewer people were needed to farm the land. People moved from the countryside to towns, to work in the newly built mills and factories. This change in the economy, from farming to manufacturing, is known as the Industrial Revolution.

COAL AND IRON

For thousands of years, iron had been made by using charcoal from timber. But in 1709 Englishman Abraham Darby and his son developed a way of making iron using coal. Coal was easier to obtain than wood, so this discovery led to a huge growth in the production of iron to make tools and machinery, and to the opening of more mines to provide the coal.



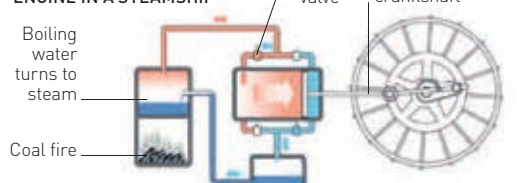
IRONBRIDGE, SHROPSHIRE

Site of the world's first iron bridge, built by Darby's grandson in 1781.

STEAM POWER

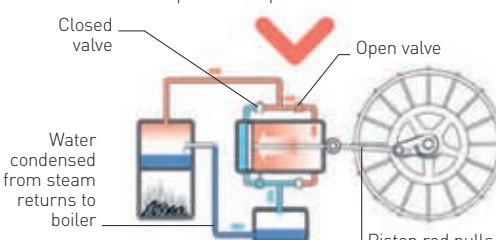
The first steam engine was invented in 1712, to suck water from mines. Then, 60 years later, James Watt developed a way of using steam to drive factory machines. The age of steam had begun and would transform industry, travel, and home life, first in Britain and then all over the world.

DOUBLE-ACTING ENGINE IN A STEAMSHIP



1 FORWARD STROKE

Steam enters the cylinder through the open valve on the left and pushes the piston forwards.



2 BACKWARD STROKE

The steam now enters the cylinder through the valve on the right and pushes the piston back.

CANALS

The growth in manufacturing meant there were more goods to be transported around the country, and canals were the answer – man-made rivers, deep enough to cope with large, heavy barges. In 1761 an engineer called James Brindley built a 16 km (10 mile) long canal to carry coal direct from a mine in Lancashire to sell in Manchester in the UK. The venture made a fortune for the mine owner and sparked a boom in canal-building.



TRENT AND MERSEY CANAL

Originally called the Grand Trunk Canal, it cost £296,600 to build and opened in 1777.



CAEN HILL LOCKS

A series of 29 locks on the Kennet and Avon Canal in England were built in 1810 by John Rennie.

BIRTH OF THE RAILWAYS

Factories needed a constant supply of raw materials coming in and finished goods going out, and the railway was the perfect solution. Trains were much quicker than canal barges, and could carry much larger quantities of goods than road wagons. By the 1840s, fast and reliable steam engines were produced and became a symbol of the Industrial Revolution across Europe and America.

STEPHENSON'S ROCKET

The *Rocket*, built by George Stephenson and his son Robert, was the first significant public steam passenger locomotive, built in 1829 to run on the new railway from Liverpool to Manchester.



TREVITHICK'S LOCOMOTIVE

In 1808 Richard Trevithick built a "rail circus" in London and charged a shilling for a ride on *Catch Me Who Can*.

